

POLARIS DC-1 — TECHNICAL SPECIFICATION

SPEC-03: ENVELOPE, CLADDING AND ROOFING

Document Ref: PDC-SPEC-03 Rev A
Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Date: 26 February 2026
Status: For Subcontract

Synthetic demonstration data. All parties, values and figures in this document are fictional.

1. Scope

- 1.1 This specification covers the insulated wall build-up (outer leaf and insulation, applied to the precast inner leaf specified in PDC-SPEC-02), the insulated roof build-up, personnel and loading doors, louvres, and the airtightness and acoustic performance of the building envelope, across all buildings on the 14,450 m² GFA site.
- 1.2 The envelope shall achieve the performance values in PDC-ER-001 §2.3 without exception: wall U-value 0.17 W/m²K, roof U-value 0.09 W/m²K, doors 1.0 W/m²K, airtightness q50 ≤ 1.0 m³/(h·m²), and shall not compromise the 45 dB(A) night-time noise limit at the north-east residential boundary.
- 1.3 This specification excludes the precast concrete inner leaf and its fire compartmentation performance (PDC-SPEC-02) and internal finishes (PDC-SPEC-04).

2. Standards and References

Reference	Subject
SFS-EN ISO 6946	Building components — thermal resistance and transmittance calculation
SFS-EN ISO 9972	Thermal performance of buildings — determination of air permeability (blower-door method)
National Building Code of Finland, Ministry of the Environment energy decree (Ympäristöministeriön asetus)	Envelope energy performance requirements
SFS-EN 14782, SFS-EN 10346	Profiled metal sheet cladding; continuously hot-dip coated steel
SFS-EN 13956	Flexible sheets for waterproofing — single-ply roof membranes
SFS-EN 13501-1	Reaction to fire classification of construction products
SFS-EN 1991-1-3, SFS-EN 1991-1-4, Finnish National Annexes	Snow and wind actions, Vantaa location
SFS-EN ISO 717-1	Rating of airborne sound insulation — building elements
SFS 4933 / RT-guidance on external noise from construction sites and plant	Acoustic assessment methodology at sensitive receptors

3. Materials

3.1 Wall build-up (achieving $U = 0.17 \text{ W/m}^2\text{K}$)

Layer	Specification
Inner leaf	150 mm precast concrete (PDC-SPEC-02)
Insulation	200 mm PIR board, thermal conductivity $\lambda \approx 0.022 \text{ W/mK}$
Cavity	50 mm ventilated cavity
Outer leaf	Profiled metal rainscreen, factory-finished, colour to the Employer's approval
Vapour control layer	Continuous, taped and lapped, positioned warm side of the insulation

3.2 Roof build-up (achieving $U = 0.09 \text{ W/m}^2\text{K}$)

Layer	Specification
Structural deck	Steel profiled deck on primary/secondary steel (PDC-SPEC-02)
Vapour control layer	Continuous, taped and lapped
Insulation	300 mm PIR board, thermal conductivity $\lambda \approx 0.022 \text{ W/mK}$, with a reinforced fixing zone at 150 mm centres coordinated for future PV mounting rails
Waterproofing	Single-ply membrane, mechanically fixed or fully adhered, hot-air welded laps
Loading allowance	Snow/wind per Finnish national annex, plus 0.75 kN/m^2 uniform PV/mounting-rail allowance, plus point loads at rooftop plant curbs

3.3 Doors and louvres

Item	Specification
Personnel and sectional loading doors	Insulated, $U \leq 1.0 \text{ W/m}^2\text{K}$, air permeability class per SFS-EN 12207 as scheduled
Louvres (cooling plant intake/exhaust)	Weather- and bird-resistant, free-area and pressure drop coordinated with the mechanical intake velocities (refer M&E specifications, outside this document's scope)
Cladding and roof fasteners	Corrosion-resistance class suited to a coastal/near-marine Finnish exposure

4. Workmanship and Execution

4.1 Air barrier continuity. The vapour control layer/air barrier at wall and roof shall be continuous across all junctions, including wall-to-roof, wall-to-slab, door and louver openings, and every structural or M&E penetration. All laps shall be taped per the manufacturer's system and inspected before being concealed by following trades (Section 6).

4.2 Panel and cladding joints. External rainscreen joints shall be sealed and drained consistently with the precast panel joint module (3.0 m centres, PDC-SPEC-02 §3) so that cladding joints and precast joints align at shared vertical lines.

4.3 Membrane installation. Single-ply roof membrane shall be installed by an installer approved by the membrane manufacturer, with all laps hot-air welded and no cold-lap joints permitted. Flashings at upstands, curbs and penetrations shall be formed before the membrane is handed over for follow-on plant installation.

4.4 **Thermal bridging.** Structural penetrations of the insulation layer (brackets, fixings, plant curbs) shall use thermal-break details at every location, coordinated at shop-drawing stage with PDC-SPEC-02.

4.5 **Sequencing against blasting and Section handover.** Envelope works on Hall 4 and the generator compound (east zone) shall not commence until the blasting works described in PDC-SPEC-01 §4.3 in the immediate work area are complete, and envelope works adjacent to an operating Section shall follow the live-environment method statement referenced in PDC-ER-001 §7.

5. Tolerances

Item	Tolerance
Rainscreen panel alignment (line and level)	± 5 mm
Rainscreen panel flatness (out-of-plane)	3 mm over 1 m straightedge
Cladding/precast joint width	15 mm ± 5 mm
Roof falls	Design fall ± 1:200, no ponding retained after 24 hours
Membrane lap width	Minimum 50 mm, per manufacturer's system
Door frame plumb and square	± 3 mm over frame height/width

6. Testing and Inspection

Test / inspection	Frequency / trigger	Acceptance basis
Air barrier lap inspection	Progressive, before concealment, 100% coverage	No unsealed laps or penetrations; photographic record retained
Whole-building/per-Section airtightness test	On completion of each Section's envelope, before internal fit-out closes access to junctions	$q_{50} \leq 1.0 \text{ m}^3/(\text{h} \cdot \text{m}^2)$, per SFS-EN ISO 9972
Membrane seam testing	100% of welded laps, vacuum box or seam probe	No leaks or unfused sections
Water penetration test (spray test at high-risk junctions)	Sample locations, minimum 3 per Section	No water ingress over test duration
U-value verification	Calculation per SFS-EN ISO 6946 at design stage; as-built spot check of insulation thickness/continuity	Calculated U-value ≤ specified value
Acoustic measurement at north-east boundary	Night-time, on completion of each Section and again at full site completion	≤ 45 dB(A), per the assessment methodology in Section 2

7. Submittals

Submission	Timing
Envelope build-up calculations (U-value, condensation risk)	S2 detailed design
Mock-up panel/junction sample, including a typical precast-to-cladding joint	Before main cladding order is placed
Air barrier detailing drawings, all junction types	S3 construction issue
Membrane manufacturer's installer approval and system warranty terms	Before roofing works commence
Airtightness test method statement	21 days before the first Section test
Acoustic assessment report demonstrating compliance at the north-east	S2 detailed design, confirmed at Section

boundary	completion
Airtightness and acoustic test results	Within 10 working days of each test

8. Spare Parts and Warranties

8.1 Single-ply roof membrane shall carry a manufacturer's system warranty of not less than 20 years; wall and roof PIR insulation shall carry the manufacturer's standard product warranty.

8.2 The Contractor shall hand over spare rainscreen cladding sheets and roof membrane offcuts sufficient for future repairs (minimum 2% of installed area of each product), together with the manufacturer's maintenance and cleaning instructions, at each Section's O&M handover.

End of PDC-SPEC-03 Rev A.

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